
Education

09/2020–06/2024 **Bachelor of Science in Biological Sciences**, Wuhan University, Wuhan City, Hubei Province, China, GPA: 3.80/4.00

Research Interests

Protein Dynamics I study how the physics of proteins—particularly the interplay of entropy and enthalpy—governs their dynamic behavior, from the thermodynamic origin of nucleation barriers in self-assembly (e.g., death fold domain filaments) to the conformational dynamics that shape molecular recognition and signaling (e.g., TCR–pMHC engagement). More broadly, I am interested in how sequence and structure encode conformational ensembles rather than single static states, and how these ensembles translate into biological function. I aim to bridge static structure prediction and dynamic ensembles by combining molecular dynamics, residue-network analysis, and machine learning to model and ultimately predict protein dynamic landscapes.

Publications

- Under Review A. N. Romantsova, O. M. Subach, S. Zhou, ..., **Y. Ling**, ..., Z. Fu, K. D. Piatkevich, F. V. Subach. *LSSmScarlet4: A Red-Shifted Large Stokes Shift Fluorescent Protein for Multicolor Imaging*.
- In Preparation X. Yin, **Y. Ling**, ..., W. Liu. *Deep mutational scanning-guided T cell receptor engineering and functional candidate discovery*.
- In Preparation X. Yin, **Y. Ling**, ..., W. Liu. *Decoding T cell receptor function through multi-dimensional mutational landscapes*.

Research Experience

03/2026–now **Research Assistant**, ZJU-UoE Institute, Jiaxing City, Zhejiang, China

Lab: Wanlu Liu Lab

Supervisor: Wanlu Liu (wanluliu@intl.zju.edu.cn)

Lab Page: Wanlu Liu Lab

Project 1: Systematic characterization of binding and activation decoupling sites on TCR

Objectives:

- Investigate binding–activation decoupling in the 1G4 TCR–NY-ESO-9C/HLA-A2 system, in which affinity-attenuating mutations preserve or enhance T cell activation despite reduced pMHC binding.
- Characterize how wild-type and decoupling-mutant TCR–pMHC interfaces differ in molecular interaction and binding dynamics across antigen-affinity variants, using atomistic molecular dynamics simulations.

Project 2: npTCR-seqHT — an ultra-high-throughput droplet platform for natively paired single-cell TCR α/β sequencing

Objectives:

- Develop in-droplet overlap-extension RT-PCR chemistry to generate natively paired TCR $\beta:\alpha$ fusion amplicons from single cells under one-pot co-encapsulation.
- Redesign the multiplexed V-gene primer panel and inter-chain linker to suppress primer-dimer and single-chain byproducts.
- Engineer a multi-stage PCR program exploiting strand asymmetry and primer-competition geometry to bias amplification toward the fusion product.

09/2024–09/2025 **Research Technician**, Stowers Institute for Medical Research, Kansas City, MO, USA

Lab: Halfmann Lab

Supervisor: Randal Halfmann (rhn@stowers.org)

Lab Page: Halfmann Lab

Project: Probing the origin and mechanism of nucleation barriers in death fold domains (DFDs)

Key Contributions:

- Identified interface hotspots encoding nucleation barriers by performing Rosetta-based saturation mutagenesis ($\Delta\Delta G$) across all three filament interface types in three DFD systems (ASC-PYD, BCL10-CARD, AIM2-PYD).
- Developed a residue-network analysis pipeline based on adjacency and communicability matrices, revealing that subunit incorporation proceeds through local network reorganization rather than global conformational change.
- Quantified the conformational entropy cost of assembly by running all-atom MD simulations of DFD monomers and filaments and computing per-residue order parameters; showed that assembly incurs a net entropic penalty despite energetically favorable interfaces.
- Established that the resulting entropic cost ranks BCL10-CARD > ASC-PYD > AIM2-PYD, consistent with their experimentally observed nucleation barriers—supporting a model in which nucleation barriers are sequence-encoded through hidden entropic costs.

- 05/2024–
08/2024 **Summer Research**, *Westlake University*, Hangzhou City, Zhejiang Province, China
Lab: Piatkevich Lab
Supervisor: Kiryl D. Piatkevich (kiryl.piatkevich@westlake.edu.cn)
Lab Page: Piatkevich Lab
Projects:
 - **LSSmScarlet4 Characterization:** Measured in vivo brightness, photostability, and chemical stability of LSSmScarlet4.
 - **Calypso Development:** Engineered a blue fluorescent protein from RFP through random mutagenesis and screening.
- 08/2023–
05/2024 **Undergraduate Researcher**, *Stowers Institute for Medical Research*, Kansas City, MO, USA
Lab: Halfmann Lab
Supervisor: Randal Halfmann (rhn@stowers.org)
Lab Page: Halfmann Lab
Project: The role of nucleation barrier in AIM2 inflammasome and CBM complex assembly
Objective: Illustrate the relationship between supersaturation level, stimulation, and response intensity in protein aggregation.
- 04/2022–
06/2023 **Undergraduate Researcher**, *Wuhan University*, Wuhan City, Hubei Province, China
Lab: National Key Laboratory of Hybrid Rice
Supervisor: Xiangsong Chen (chen.xs@whu.edu.cn)
Project: Research and application of epigenetic modifications in high-temperature and salt-tolerant hybrid rice varieties (04/2022–09/2022).

Competitions

- 01/2022–
11/2022 **International Genetically Engineered Machine Competition (iGEM)**, *Wuhan University Team*, Wuhan City, Hubei Province, China
Project: A Novel Anti-Depression Therapy (Project Link)
Objectives: Design a probiotic-based system with modules for production, oscillation, directed evolution, and safety switch to secrete antidepressant SAME via *Bacillus subtilis* and *E. coli*.
Achievements:
 - Engineered SAME synthesis and secretion mechanisms in *B. subtilis*.
 - Developed a REPRESSILATOR oscillator for periodic SAME release.
 - Implemented directed evolution for lactose-tolerant LacI repressors.
 - Designed safety switch modules for controlled biocontainment.

Talks

- 07/01/2025 **Ling, Y.**; Hettige, J.; Halfmann, R. “Why DON’T DFDs just assemble?” 30th FASEB Protein Aggregation Symposium, Arizona, USA.

Posters

- 06/28/2025–
07/02/2025 **Ling, Y.**; Hettige, J.; Halfmann, R. “Probing the Thermodynamic Origins of Nucleation Barrier in DFDs” 30th FASEB Protein Aggregation Symposium, Arizona, USA.
- 01/18/2025–
01/22/2025 **Ling, Y.**; Miller, T.; Song, X.; Halfmann, R. “Sequence-encoded nucleation barriers govern adaptor protein self-assembly” 30th Anniversary Midwest Stress Response and Molecular Chaperone Meeting, Chicago, USA.

Technical Skills

- Dry lab skills** High-Performance Computing (HPC), Linux, Python, Molecular Dynamics Simulation (GROMACS, AMBER), Residue-network analysis, Data analysis
- Wet lab skills** Cell experiments: cell freeze, cell recovery, adherent and suspension cell culture, transfection, transduction; Yeast experiments: yeast culture, transformation; Mouse experiments: injection, blood collection, cliff test, primary T cell collection; Molecular experiment: PCR, plasmid construction, transformation, electrophoresis, western blot; Flow cytometry; Imaging

Languages

- English Fluent (TOEFL 5.5/6.0, 110/120)
Chinese Native